

Database Triggers (Self-Learning)

Q. What is a Trigger in SQL? Discuss the different types of triggers (BEFORE and AFTER) and explain their primary uses.

Definition to Remember

1. Types of Triggers

Triggers are classified by **when** they execute relative to the triggering DML event:

A. BEFORE Trigger

Example: Before inserting into the `Employee` table, a BEFORE trigger checks if the incoming salary is below minimum wage. If so, it **raises an error and aborts** the insert.

B. AFTER Trigger

Example: After an update to `Employee`, an AFTER trigger inserts a record into `Employee\_Audit` logging the user, timestamp, and what changed.

(Some systems also support **INSTEAD OF Triggers** primarily used on Views to enable update operations on non-updatable views.)

2. Trigger Event Classification

BEFORE | AFTER

3. Uses and Advantages

Use Case | Explanation

> **Must-Write Points (for 10 marks)**

> 1. A

> **Quick Recall**

> `Trig
Auditing